Internship Report: Reflections on an Enriching Journey with the PreProd Team

Introduction

The journey with the PreProd Team has been nothing short of transformative. Over the course of this internship, I encountered challenges that tested my capabilities, resilience, and adaptability. Each week brought new opportunities to delve into unfamiliar territories, explore innovative solutions, and expand my technical and professional acumen. While the journey was demanding, it was immensely rewarding and filled with invaluable learning experiences.

Weekly Progression and Experiences

Week 1: Introduction to Agile Framework

The first week laid the foundation for the weeks to come as we immersed ourselves in the **Agile methodology**. This framework proved to be a cornerstone for future self-directed, team-based, and organizational projects. Understanding the principles of Agile not only enhanced my collaborative skills but also improved my approach to structured problem-solving and iterative development.

Week 2: Exploring Data Streaming and Linux Frameworks

During the second week, I delved into the world of **data streaming**, where I learned and implemented Apache Kafka. This marked my first encounter with the Linux environment, which introduced me to virtual machines, Windows Subsystem for Linux (WSL), and tunneling. The learning curve was steep, but with the guidance of my mentor, Vansh, I successfully navigated these complexities.

Week 3: Mastering Tree-Based Machine Learning Models

The third week focused **on tree-based machine learning models, hyperparameter tuning, and data preprocessing**. I worked extensively with diverse datasets and familiarized myself with **PySpark**, which added significant depth to my data processing toolkit. Vansh once again provided valuable support, ensuring that I could overcome any roadblocks.

Week 4: Comprehensive Spike Ticket Development

In the fourth week, I contributed to the team's spike ticket project, which encompassed data versioning, model building, tracking, and serving. This experience introduced me to tools like **MLFlow, Dagshub, and DVC**, allowing me to understand the lifecycle of data-driven projects comprehensively. The collaborative effort highlighted the importance of teamwork and systematic execution.

Week 5: Challenges of Reinforcement Learning

The fifth week presented the most intellectually demanding challenge: reinforcement learning. My task involved developing a **Deep Q-Network (DQN) to address the Cart-Pole problem**. Crafting policies, rewards, and penalties to train the model required persistent effort and a deep understanding of the underlying concepts. While I made significant progress, I acknowledge there is still more to learn in this domain.

Week 6: Advancing with YOLO and Hugging Face Library

Week six allowed me to explore **YOLO for a dog breed detection model** and the **Hugging Face library**. I experimented with multiple NLP models for tasks like sequence classification, text generation, sentiment analysis, question-answering, and text summarization. The experience of using models such as **DistilBERT, GPT-2, RoBERTa, and T5** was enlightening. Additionally, I developed a module to scrape web reviews and compute sentiment-based ratings, blending creativity with technical expertise.

Week 7: Individual Project on Amazon Review Analysis

In the final week, I undertook an individual project under the mentorship of Krishnav Sir, focusing on the Amazon Review Problem. This project involved exploring text representation techniques like **TF-IDF, Bag of Words, Word2Vec, and Averaged Word2Vec**. Moreover, I had the opportunity to **fine-tune a DistilBERT model**, marking a significant milestone in my journey with NLP. Unfortunately, health issues caused some delays, but the overall learning was immensely satisfying.

Acknowledgements

This internship would not have been as fruitful without the unwavering support and guidance of my mentors and teammates. Vansh's expertise and patience were instrumental in overcoming technical hurdles. Harish Sir, Krishnav Sir, and Mohini provided invaluable insights and encouragement throughout the journey. Their mentorship fostered a culture of learning, accountability, and collaboration.

Conclusion

Although the internship concluded with unforeseen challenges, the overall experience has been transformative. The knowledge, skills, and memories gained during these seven weeks have left an indelible mark on my professional and personal growth. I hope to carry forward the lessons learned and look forward to future opportunities to collaborate with the PreProd Team.